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A Growing Visible Light Leak in the 94 Å Channel of the Atmospheric Imaging Assembly

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ABSTRACT

The 94 Å channel of the Atmospheric Imaging Assembly (AIA) on the Solar Dynamics Observatory (SDO) is the instrument's main view of coronal plasma hotter than a few megakelvin, and with the other extreme ultraviolet (EUV) channels it underlies most differential emission measure (DEM) analyses of the corona. We find that the channel also records visible light from the photosphere. The leak is a sharp, limb-darkened image of the solar disk with a broad halo, and since each visible photon frees a single electron, it adds almost none of the photon noise that an EUV signal of the same size would. It is absent from the 304 Å channel, which shares the telescope and the detector, it is attenuated more than the EUV by the thicker of the two focal-plane filters, and it can be seen directly as SDO passes into the shadow of the Earth, where it sets 76 s after the EUV has been absorbed, flattened by refraction like a sunset. In 28 such eclipses between 2010 and 2026, the leak at disk center grew from 0.032 DN s^{-1} to 0.88 DN s^{-1}, exponentially since 2016 with a doubling time of 3.1 yr, and it is now 66% of the quiet-Sun signal in the channel. Cross-calibration with the Solar Ultraviolet Imager (SUVI) on GOES-18 and GOES-19 and the noise of the channel itself give the same history. Left in the data, the leak appears in DEMs as hot plasma in the quiet Sun and in coronal holes, it biases maps of Fe XVIII, and it hides part of the degradation of the channel from its calibration. We give a model of the leak and its history from which it can be subtracted.

Keywords: Solar instruments (1499), Calibration (2179), Solar extreme ultraviolet emission (1493), Solar corona (1483), Quiet sun (1322), Solar coronal holes (1484), Eclipses (442), Limb darkening (922)

1. INTRODUCTION

Outline.

- **The channel.** The Atmospheric Imaging Assembly (AIA) (J. R. Lemen et al. 2012) images the corona in seven extreme ultraviolet (EUV) channels from the Solar Dynamics Observatory (SDO) (W. D. Pesnell et al. 2012). The 94 Å channel sees Fe XVIII from plasma near 7 MK, and cooler lines of Fe IX and Fe X which atomic databases underestimate (P. Testa et al. 2012), so its temperature response was corrected empirically against the EUV Variability Experiment (EVE) (T. N. Woods et al. 2012; P. F. Boerner et al. 2014).
- **What depends on it.** It is the main constraint on hot plasma in differential emission measure (DEM) inversions (I. G. Hannah & E. P. Kontar 2012; J. Plowman et al. 2013; M. C. M. Cheung et al. 2015; H. Morgan & J. Pickering 2019; J. Plowman & A. Caspi 2020), and maps of Fe XVIII are made by subtracting the cool part of it predicted from the other channels (H. P. Warren et al. 2012; G. Del Zanna 2013).
- **Calibration over the mission.** The degradation of each channel is tracked against EVE and its successors (P. Boerner et al. 2012; P. F. Boerner et al. 2014; L. F. G. Dos Santos et al. 2021) and distributed in `aiapy` (W. Barnes et al. 2020). A signal that grows with time reads as a channel which degrades more slowly than it does.
- **A precedent.** The zirconium filters of the 94 Å channels transmit a little visible light (S. K. Tadikonda et al. 2019), and pinholes in the entrance filters of the Solar Ultraviolet Imager (SUVI) on GOES-17 leak visible light, most in its 94 Å channel (J. M. Darnel et al. 2022). Check that Darnel2022 describes the GOES-17 leak; otherwise cite the SUVI L1b readme.
- **Symptoms already in the literature.** In coronal holes the 94 Å signal matches what remains behind an occulter, and was taken for stray light (J. Saqri et al. 2020; S. G. Heinemann et al. 2021). A survey of quiet-Sun DEMs ended in 2018 because the reconstructions after it were poor (S. Purkhart & A. M. Veronig 2022), and some inversions drop the channel (S. J. Hofmeister et al. 2025). Coronal holes lose their contrast in it (S. M. Lezzi et al. 2023), and it is brighter than a

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DEM constrained in X-rays allows (P. S. Athiray & A. R. Winebarger 2024).

- **Roadmap.** Section 2 finds a noiseless signal in the channel, Section 3 measures it directly in eclipses over the mission, Section 4 confirms it with the focal-plane filters, the other channels, and SUVI, Section 5 follows it into thermal diagnostics and calibration, Section 6 gives a correction, and Section 7 concludes.

2. PHOTON NOISE

Outline.

- **Data.** A run of 15 consecutive level 1 frames at noon on each day of an eclipse, returned to the units of the detector by multiplying each by the flat field it was divided by.
- **Method.** The residual of each frame from the mean of its two neighbors cancels the Sun and anything that changes linearly with time, and has 3/2 the variance of one frame. It is sorted by the signal of the other frames of the run, whose noise is independent of it. The read noise, about 1.2 DN^2, comes from the corners of the charge-coupled device (CCD) beyond the field stop; each quadrant of the CCD is read out by its own amplifier, and is fit on its own.
- **The noiseless signal.** Off the limb the variance rises from the read noise with the signal, as photon noise alone would, at the slope found near disk center. Near disk center the line of slope 0.54 DN meets the read noise only at a signal of 1.9 DN$\pm$0.2 DN: that much of the signal carries no photon noise (Figure 1, left). The photon noise of a signal grows with the electrons each photon frees in the silicon: dozens for a 94 Å photon, one for a visible photon, so a signal of visible light carries almost none.
- **Its map.** The noiseless signal fills a disk the size of the photosphere (Figure 1, middle), and does not follow the corona: across the disk it grows by 0.02 per DN of the 94 Å signal of a region, though it is 63% of that signal. It darkens toward the limb as $1 - u(1 - \mu)$ with $u = 0.57$ (Figure 1, right), where $\mu = \sqrt{1 - r^2}$ is the cosine of the angle between the line of sight and the vertical. This is limb darkening, the signature of an optically thick photosphere: we see into it only to a fixed optical depth along the line of sight, which near the limb, at a grazing angle, lies in higher and cooler layers, so the disk dims toward its edge. The optically thin corona does the opposite, brightening toward the limb, where the line of sight through it is longest. The channel sees the photosphere, in light that frees few electrons per photon: a visible light leak. Cite a measured limb darkening of the photosphere in visible light (e.g., Neckel and Labs 1994) to compare u with.

3. EARTH ECLIPSES

Outline.

- **Eclipse seasons.** SDO passes behind the Earth once a day for a few weeks twice a year, and the seasons drift earlier from year to year.
- **Ingress.** The atmosphere absorbs the EUV high above the surface, while the visible disk sets only near it, so for a while each part of the disk is seen in visible light alone. The 171 Å channel marks where the EUV has gone (Figure 2).
- **Method.** Each bin of the disk on a clock started by its own 171 Å edge; a forward model of a limb-darkened disk setting behind the atmosphere, seen through a point spread function (PSF) with a sharp core and broad wings, plus whatever EUV is left, fit to the frames in which the EUV is gone. The frames in the umbra are the dark.
- **Results.** The leak at disk center in 28 eclipses from 2010 to 2026: from 0.032 DN s^{-1} to 0.88 DN s^{-1}, with a doubling time of 3.1 yr since 2016 (Figure 3). Its limb darkening, the sharpness of its limb, and the halo beyond it.
- **Against the noise.** The noiseless signal near disk center in the run at noon of each day follows the leak (Figure 3). Since 2019, when the leak first exceeded three times the spread of the noise method among the quadrants (0.1 DN), the noiseless signal is 72% to 100% of the leak, the fraction falling as the leak grows; before, the two differ by up to a factor of two. Part of the shortfall is expected: a visible photon frees one electron, so the leak does carry a little photon noise, a twentieth of what an EUV signal of the same size would. Account for the rest: the rise of the variance per DN with signal, the step in the noiseless signal between the quadrants, and the change of the read noise in 2018.
- **Caveats.** The PSF is fit once to the brightest eclipses and held fixed, since in a single eclipse its core and wing trade off against each other, though the light they deliver to the disk does not. Before 2016 the leak is too faint to time its setting, which is held at its median since, and it lies above the exponential the later eclipses follow. One ingress fell during a flare.
- **Figures.** Still to come: the radial profile of the leak, and the sharpness of its limb.

4. INDEPENDENT CHECKS

Outline.

- **The focal-plane filters.** On the one day since 2024 that the channel observed through its thick focal-plane filter, the leak fell by more than the EUV did, so it passes through that filter and is light rather than an artifact of the detector or its calibration.
- **The other channels.** The 304 Å channel shares the telescope and the CCD but has no leak, so the leak is in the filters of the 94 Å half of the aperture. What the noise says, and cannot say, about the 131, 171, 193, 211, and 335 Å channels.

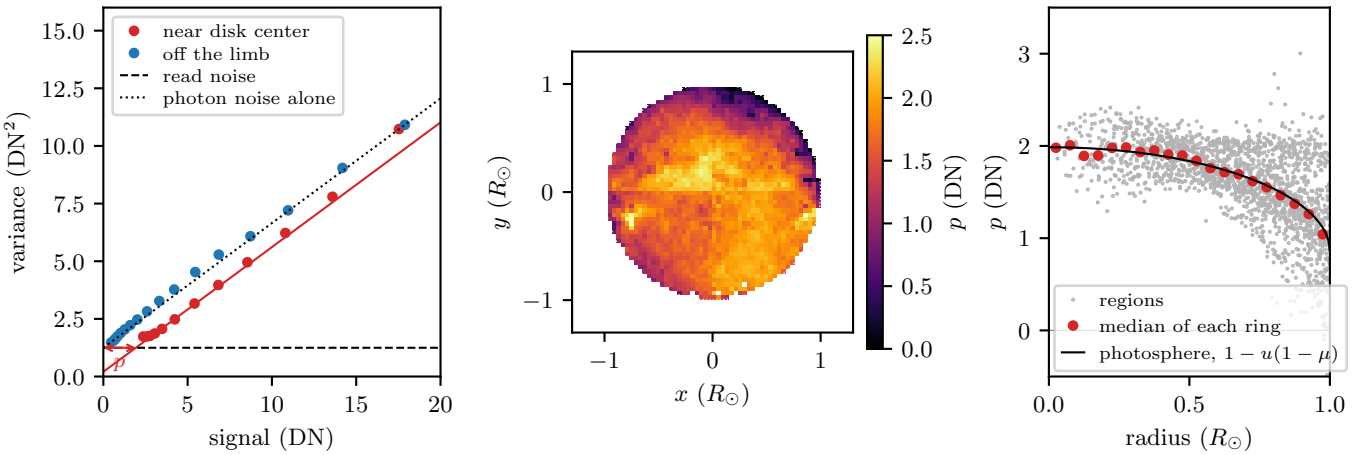


Figure 1. The noise of 94 Å in a run of 15 frames at noon on 2026 July 15, in the units of the detector. Left: the variance of one frame against its signal, near disk center (within half a radius) and off the limb (1.05 to 1.3 radii), and the read noise from the corners beyond the field stop. Off the limb the variance rises from the read noise; near disk center it reaches the read noise only at a signal of $p = 1.9$ DN $\pm$ 0.2 DN, which carries no photon noise. The dotted line is the variance if all of the signal carried photon noise, the slope near disk center started from the read noise at zero signal: the points off the limb follow it, and those near disk center lie p to its right. Each quadrant of the detector, read out by its own amplifier, is fit with its own read noise and slope; the line and the read noise shown are their means, and the uncertainty of p is its spread among the quadrants. Middle: p in each region of 64 × 64 pixels on the disk, with the read noise and the slope of its quadrant; off the limb the slope of the disk does not hold. Right: its radial profile, each region (gray) and the median of each ring of 0.05 $R_\odot$ (red), with the limb darkening of a photosphere, $1 - u(1 - \mu)$, where $\mu = \sqrt{1 - r^2}$ is the cosine of the angle between the line of sight and the vertical, fit to the regions within 0.95 $R_\odot$: $u = 0.57$.

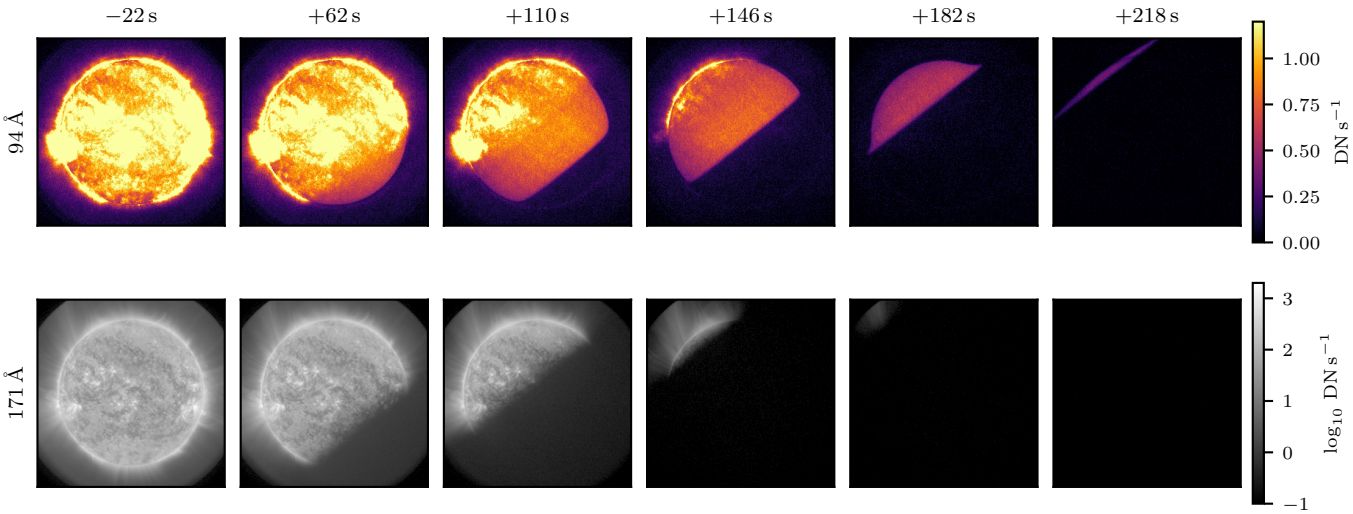


Figure 2. The ingress of the eclipse of 2026 July 15 in the 94 Å (top) and 171 Å (bottom) channels, at the times marked after the 171 Å disk mean began to fall. Each frame is averaged over 16 × 16 pixels, less the umbra, and divided by its exposure time; 94 Å is in the units of the detector. The atmosphere absorbs the EUV from the lower right, and the corona of both channels goes with it, but in 94 Å a smooth, limb-darkened disk is left behind. It sets 81 s after the EUV edge has passed, flattened by refraction, as the visible Sun does.

- **SUVI.** A joint fit of AIA to SUVI on GOES-16, GOES-18, and GOES-19 with a limb-darkened term on the disk and a constant gives the same history, the constant being the halo.

5. CONSEQUENCES

Outline.

- **Differential emission measures.** The fraction of the DEM of coronal holes and the quiet Sun above 3 MK, early in the mission and now, with and without the leak subtracted. Subtracting it restores the quiet

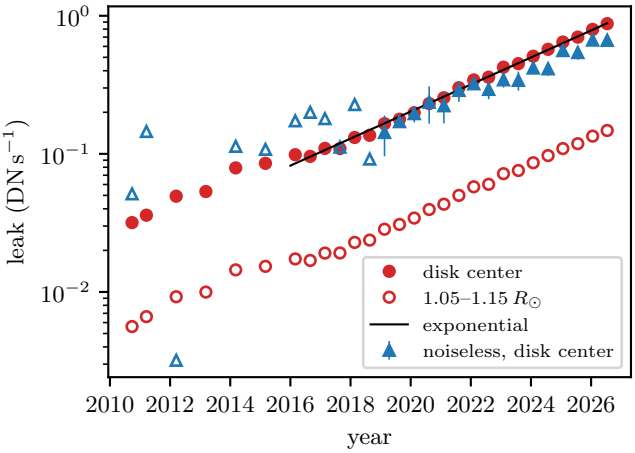


Figure 3. The leak in the 28 eclipse ingresses, in the units of the detector, at disk center (filled circles) and just off the limb (open circles), each with the whole disk up, and the noiseless signal near disk center in a run at noon of each day (triangles), with its spread among the quadrants of the detector; it is open where the leak is less than three times the typical spread, 0.1 DN, and the noise does not measure it. The line is an exponential fit to the eclipses since 2016, which doubles every 3.1 yr.

Sun, but coronal holes stay hotter than they were early in the mission.

- **Fe XVIII.** In quiet regions the leak survives the subtraction of the cool part of the channel.
- **Calibration.** In full-disk comparisons a growing leak reads as rising sensitivity. The correction tables now hold the sensitivity of the channel constant.
- **Published results.** Which of the results of Section 1 the leak accounts for, and how much.

6. A CORRECTION

Outline.

- **Model.** A limb-darkened disk seen through the PSF of the leak, scaled by an amplitude interpolated between eclipses, in the units of the detector and divided by the flat field.
- **Uncertainty.** From the scatter about the exponential and between the methods.
- **Keeping it current.** Two eclipse seasons a year measure the leak as it grows. Decide where the correction lives: `aiapy`, `sdo`, or this package.

7. CONCLUSIONS

Outline.

- The 94 Å channel records a growing image of the photosphere in visible light.
- Four independent measurements agree on its size, shape, and history.
- What it means for hot plasma outside active regions since about 2018, and what to do about it.
- Where the leak enters the instrument, and how to tell, for the AIA team.

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